Probability as a number

Core Concepts: Probability as a number

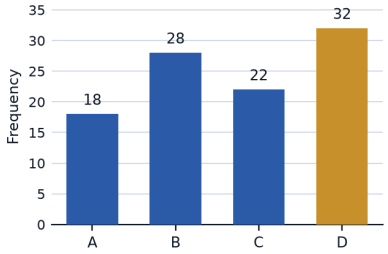
1 For the model $f(x) = 4x + 6$ in probability as a number, evaluate $f(2)$.

a. 14

b. 18

c. 8

d. 18



2 Find the positive solutions to the quadratic equation $x^2 - 4x + 3 = 0$.

a. $x = 1$ and $x = 4$

b. $x = 1$ and $x = 3$

c. $x = -1$ and $x = -3$

d. $x = 4$ and $x = 3$

3 A data set recorded for probability as a number consists of the four values $\{13, 17, 21, 27\}$. What is the arithmetic mean?

a. 17.5

b. 18.5

c. 19.5

d. 21.5

Procedural Fluency & Algebraic Manipulation

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4 For the model $f(x) = 4x + 6$ in probability as a number, evaluate $f(3)$.

a. 18

b. 22

c. 12

d. 22

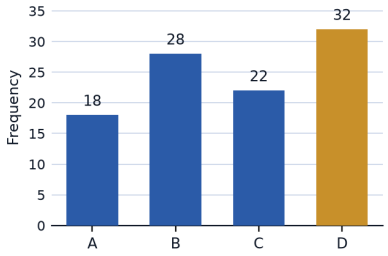
5 Find the positive solutions to the quadratic equation $x^2 - 6x + 8 = 0$.

a. $x = 2$ and $x = 5$

b. $x = 2$ and $x = 4$

c. $x = -2$ and $x = -4$

d. $x = 6$ and $x = 8$



6 A data set recorded for probability as a number consists of the four values $\{13, 17, 21, 27\}$. What is the arithmetic mean?

a. 17.5

b. 18.5

c. 19.5

d. 21.5

Graphical & Geometric Interpretation

7 For the model $f(x) = 4x + 6$ in probability as a number, evaluate $f(4)$.

a. 22

b. 26

c. 16

d. 26

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8 Find the positive solutions to the quadratic equation $x^2 - 8x + 15 = 0$.

a. $x = 3$ and $x = 6$

b. $x = 3$ and $x = 5$

c. $x = -3$ and $x = -5$

d. $x = 8$ and $x = 15$

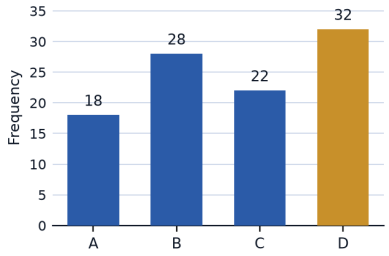
9 A data set recorded for probability as a number consists of the four values $\{13, 17, 21, 27\}$. What is the arithmetic mean?

a. 17.5

b. 18.5

c. 19.5

d. 21.5



Exam-Style Problem Solving & Modeling

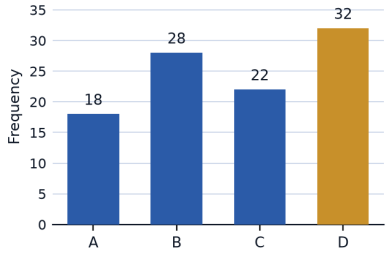
10 For the model $f(x) = 4x + 6$ in probability as a number, evaluate $f(5)$.

a. 26

b. 30

c. 20

d. 30



11 Find the positive solutions to the quadratic equation $x^2 - 10x + 24 = 0$.

a. $x = 4$ and $x = 7$

b. $x = 4$ and $x = 6$

c. $x = -4$ and $x = -6$

d. $x = 10$ and $x = 24$

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12 A data set recorded for probability as a number consists of the four values $\{13, 17, 21, 27\}$. What is the arithmetic mean?

a. 17.5

b. 18.5

c. 19.5

d. 21.5

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